frontiers in Education

TYPE Systematic Review
PUBLISHED 26 March 2026
DOI 10.3389/feduc.2026.1561649

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OPEN ACCESS

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RECEIVED 16 January 2025
REVISED 10 February 2026
ACCEPTED 03 March 2026
PUBLISHED 26 March 2026

CITATION
Putra ZH, Alim JA, Kurnia R, Gunawan Y, Dahnilsyah D and Aljarrah A (2026) A systematic literature review of joyful learning in mathematics: an implication for teaching and learning in elementary school.
Front. Educ. 11:1561649.
doi: 10.3389/feduc.2026.1561649

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A systematic literature review of joyful learning in mathematics: an implication for teaching and learning in elementary school

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Elementary Students often perceive mathematics as difficult and disengaging. Joyful mathematics learning has been progressively explored as an instructional approach to make mathematics more meaningful and enjoyable. This study presents a systematic literature review (SLR) of 30 articles published between 2017 and 2024, aiming to describe how joyful learning is conceptualized, implemented, and investigated within mathematics education. The analysis of the reviewed studies is categorized by mathematical domains, research methods, school levels, technology integration, and pedagogical approaches. The findings reveal that joyful learning is most frequently applied in geometry and arithmetic, with classroom action research emerging as the dominant methodology. Technology, student worksheets, and game-based learning are often integrated to enhance engagement. While many studies suggest positive effects on student motivation and participation, further empirical research is needed to substantiate claims regarding learning outcomes. This review provides valuable insights into current practices and highlights opportunities for future research and instructional design in joyful mathematics learning.

KEYWORDS
elementary school, joyful mathematical learning, learning mathematics, PRISMA, systematic literature review

1 Introduction

Joyful learning has emerged as an influential pedagogical approach that emphasizes creating positive emotions, intrinsic motivation, and meaningful engagement in classroom activities. It integrates elements such as games, storytelling, puzzles, simulations, and exploratory tasks, which enable students to experience enjoyment while engaging with academic content (Mishra, 2017; Wei et al., 2011). Rather than positioning enjoyment as mere entertainment, joyful learning promotes a cognitively stimulating and emotionally supportive environment in which students feel safe to take risks, collaborate, and persist in problem solving (Handayani et al., 2021; Yabo, 2020). In mathematics education, this approach has been recognized as particularly powerful: it reduces anxiety, supports active participation, and transforms abstract concepts into more accessible and relatable experiences (Anggoro et al., 2017; Cronqvist, 2021). When students feel happy and engaged, they are more likely to participate actively, persevere through challenges, and achieve higher levels of understanding (Kruk et al., 2022). Previous research further demonstrates that joyful learning fosters deeper

conceptual understanding, enhances problem-solving skills, and strengthens long-term retention of mathematical knowledge (Applebaum, 2025; Ay, 2022; Torres-Peña et al., 2025).

Despite its benefits, mathematics continues to be widely perceived as a difficult and disengaging subject by both elementary and secondary students. Studies reveal that mathematics is often regarded as complicated (Martinez and Vos, 2017) and complex (Vos, 2018), which results in students developing negative attitudes toward the subject (Ernest, 2018). Many learners remain passive, reluctant to inquire about the relevance of what they are taught, and disengaged from classroom discussions (Clarke and Roche, 2018; McCulloch et al., 2018; Putra et al., 2022). Some students require additional support to grasp fundamental concepts (Jayanti et al., 2023; Manik et al., 2022). According to Mazana et al. (2018), these problems stem from both student and teacher factors. Student-related challenges include limited comprehension of foundational ideas (Ingram et al., 2020; Kenedi et al., 2019; Namkung et al., 2019), while teacher-related issues are linked to persistent reliance on conventional, teacher-centered methods characterized by rote memorization, lecture-based delivery, and limited contextualization (Fitriana et al., 2018; Setyowati et al., 2020). Such conditions underscore the urgent need for innovative instructional strategies that move beyond traditional pedagogy and provide students with more engaging, meaningful, and enjoyable mathematical experiences.

The integration of joyful learning in mathematics is well-grounded in constructivist theory, as articulated by Piaget (1976), who emphasized that children construct knowledge through active engagement with their environment (Von Glasersfeld, 1982). This theoretical foundation explains why pedagogical approaches that involve interactive exploration and play are effective in promoting both motivation and comprehension. When mathematics instruction is designed to be joyful through activities such as collaborative games, contextualized puzzles, or culturally relevant problem solving students are not only more motivated but also develop a deeper, lasting understanding of mathematical concepts (Chen et al., 2016; Mishra, 2017). Moreover, joyful learning reshapes students' perceptions of mathematics, enabling them to view the subject not as an inherently difficult discipline but as one that is approachable, relevant, and enjoyable (Jeet and Pant, 2023; Ncube and Luneta, 2025). Given the increasing demand for student-centered learning in the 21st century, joyful learning offers a timely and relevant response to contemporary educational challenges (Kim et al., 2022).

Building upon this theoretical and empirical foundation, the present study aims to systematically review research on joyful learning in mathematics education. By analyzing 30 peer-reviewed articles published between 2017 and 2024, this review seeks to clarify how joyful learning has been conceptualized, implemented, and studied within mathematics classrooms. Specifically, it addresses six guiding research questions:

1. What mathematical domains are integrated into joyful mathematical learning?
2. What research methods are used to apply joyful mathematical learning?
3. What are the levels of schools that implement joyful mathematical learning?
4. How is technology integrated into joyful mathematical learning?
5. What are the pedagogical approaches that integrate joyful mathematical learning?
6. What are the research findings that implement joyful mathematical learning?

2 Literature review

2.1 Joyful learning

Joyful learning is an educational approach designed to develop positive emotions and engagement by incorporating amusement, creativity, and meaningful interaction into classroom activities (Mishra, 2017; Wei et al., 2011). It aims to create a cognitively stimulating environment rather than simply entertaining students, where they feel safe, supported, and motivated to take risks and persist in problem-solving tasks (Waterworth, 2020). In mathematics classrooms, for instance, Joyful learning focuses on collaborative games, visual manipulatives, storytelling, or contextual problem-solving tasks using real-world examples. A concrete example would be using the "Tarsia Puzzle" to teach geometry concepts, allowing students to work in groups while matching shapes and formulas in a game format (Palarao et al., 2025; Qomaria, 2021). This method not only enhances conceptual understanding but also promotes social interaction and reduces anxiety, which is often associated with math learning.

Numerous studies have confirmed that joyful learning enhances student engagement, creativity, and academic outcomes by cultivating intrinsic motivation (Setyaningsih and Dayu, 2022; Tafani and Kamaludin, 2023; Wicaksono, 2020). A comfortable and emotionally safe classroom has been fully evident with the practice of joyful learning in the classroom in the form of active participation, reduced math anxiety, and promotion deeper conceptual understanding (Amekauma, 2024; Li et al., 2024). For example, students who learn arithmetic through storytelling or games are more likely to retain procedural knowledge and apply it flexibly in problem-solving contexts. In line with this, several studies have also demonstrated that joyful learning strategies significantly contribute to improving student learning outcomes (Istiqomah and Prihatnani, 2019; Sufiani and Marzuki, 2021; Setyowati et al., 2020). Furthermore, when combined with technology such as Powtoon animation videos (Tafani and Kamaludin, 2023), integrated with diverse pedagogical approaches (Nathan, 2020; Wahyuni and Naim, 2019), or supported by innovative learning media (Handayani et al., 2021; Setyaningsih and Dayu, 2022), these findings indicate that joyful learning not only supports engagement but also fosters creativity, collaboration, and problem-solving skills.

Strategies for implementing joyful learning include (1) designing activities that allow opportunities for student choice and exploration, (2) utilizing engaging content such as animations, puzzles, or culturally relevant examples, and (3) maintaining instructional flexibility so teachers can adapt dynamically to classroom conditions. This flexibility, referred to

as "updates in the learning process" by Widyawulandari et al. (2019), highlights teachers' ability to adjust strategies based on formative feedback or real-time challenges, ensuring that learning remains meaningful and responsive.

2.2 Joyful mathematical learning

Problems in learning mathematics are often perceived as complex undertakings (Gunawan et al., 2023; Santos-Trigo, 2020). In the course of the instructional process, students are usually often sleepy, inattentive, and tend to be less engaged, so teachers therefore should be creative in making mathematics learning more interesting and memorable (Bukit et al., 2023). One way of achieving this is through joyful learning in mathematics, which emphasizes meaningful and enjoyable activities rather than routine memorization. Joyful learning in mathematics is a process of learning mathematics that, in its implementation, creates a sense of pleasure and comfort for students (Yabo, 2020). Beyond enjoyment, this approach fosters higher-order outcomes such as student engagement, critical thinking, and problem-solving skills by providing interactive, contextual, and playful experiences that support deeper understanding (Chen et al., 2016; Mishra, 2017).

The challenge that is usually faced in the mathematics learning process is that teachers do not sufficiently involve students in active participation (Habeahan et al., 2023), which results in students feeling bored and disengaged (Nurlaily et al., 2019; Taleb et al., 2015). The methods commonly used remain traditional, monotonous, and rarely encourage joyful learning (Doabler and Fien, 2013; Sufiani and Marzuki, 2021). For example, in multiplication lessons, teachers often provide practice problems directly without the use of learning media or games, leaving some students unable to understand how to solve them. In contrast, joyful learning can be applied through relaxed and playful activities (Ubay et al., 2021), including mathematics games (Mufidah, 2023; Waldfogel et al., 2023), which create a more pleasant learning atmosphere and encourage new ways of understanding (Khanna and Surabhi, 2020). The application of joyful learning in mathematics allows students to learn while playing, which reflects the essence of children's natural play (Keung and Fung, 2021). This aligns with Martinez and Vos (2017), who revealed that involving children in fun mathematics activities makes them more interested and motivated to participate actively.

3 Research methodology

3.1 Methods

The research method used is a Systematic Literature Review (SLR). SLR summarizes several initial studies conducted, which are then used to present comprehensive and balanced facts (Mengist et al., 2020; Pati and Lorusso, 2018). SLR is scientific research that only focuses on answering specific questions using specific scientific methods, identifying, selecting, summarizing, and evaluating similar studies in depth. In addition, it provides a deeper understanding of the research question through an extensive database of articles to be reviewed. The main reason for choosing SLR is to adopt and collect literature on joyful mathematical learning in research. The research pattern used in determining criteria and data collection uses the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA). The stages consist of (a) the identification, (b) the screening, (c) the eligibility, and (d) the quality assessment (Figure 1).

3.2 Systematic search strategy

3.2.1 Identification

Identification is the initial stage in searching literature from database sources (Van Dinter et al., 2021). There are five ways used in searching the literature from the database. The first way is to use the Publish or Perish (PoP) version 8 for Windows application to search for articles published in English in Scopus-indexed journals. The second way is to access the DOAJ portal database page (https://doaj.org/) directly, the third way is to access the ERIC portal page (https://eric.ed.gov/) directly, the fourth way is to access the Garuda portal page (https://garuda.kemdikbud.go.id), and the fifth way is to access the Sage portal page (https://journals.sagepub.com/) directly, as presented in Table 1. The reason for using the Publish or Perish (PoP) application is to make it easier for authors to search, retrieve, and analyze articles and writings contained in the database in Scopus Journals. The author also uses PoP to help select the best articles in the Scopus database. In addition to PoP, one of the reasons the author uses the DOAJ and ERIC portals is that they are trusted, reputable international journals with global open access to various disciplines. Meanwhile, the Garuda portal was used to access journals published in Indonesia that meet the criteria set by the Indonesian government. Meanwhile, researchers also searched for additional articles on the Sage portal.

The next step is to search sources using the keyword "Joyful Learning" to get access to articles in the database. The search for articles in the five databases was restricted to publications from 2017 to 2024. The literature search was limited to 2017–2024 to obtain recent developments in joyful learning aligned with Indonesia's educational reform. Since 2017, the government has promoted student-centered approaches, and in 2021, it officially introduced deep learning through the prototype curriculum, later formalized as the Merdeka Curriculum in 2022. Joyful learning serves as a base foundation toward deep learning. This time frame ensures relevance to the current policy transitions and pedagogical practice.

The results of the literature search showed 415 articles, namely 200 articles published in Scopus-indexed international journals, 72 articles published in DOAJ journals, 25 articles published in ERIC journals, 93 articles published in the Garuda portal, and 25 articles published by the Sage publisher (Figure 2).

3.2.2 Screening

The screening stage was applied through a comprehensive examination of the title, abstract, and keywords. The screening results showed 50 articles derived from the identification, while

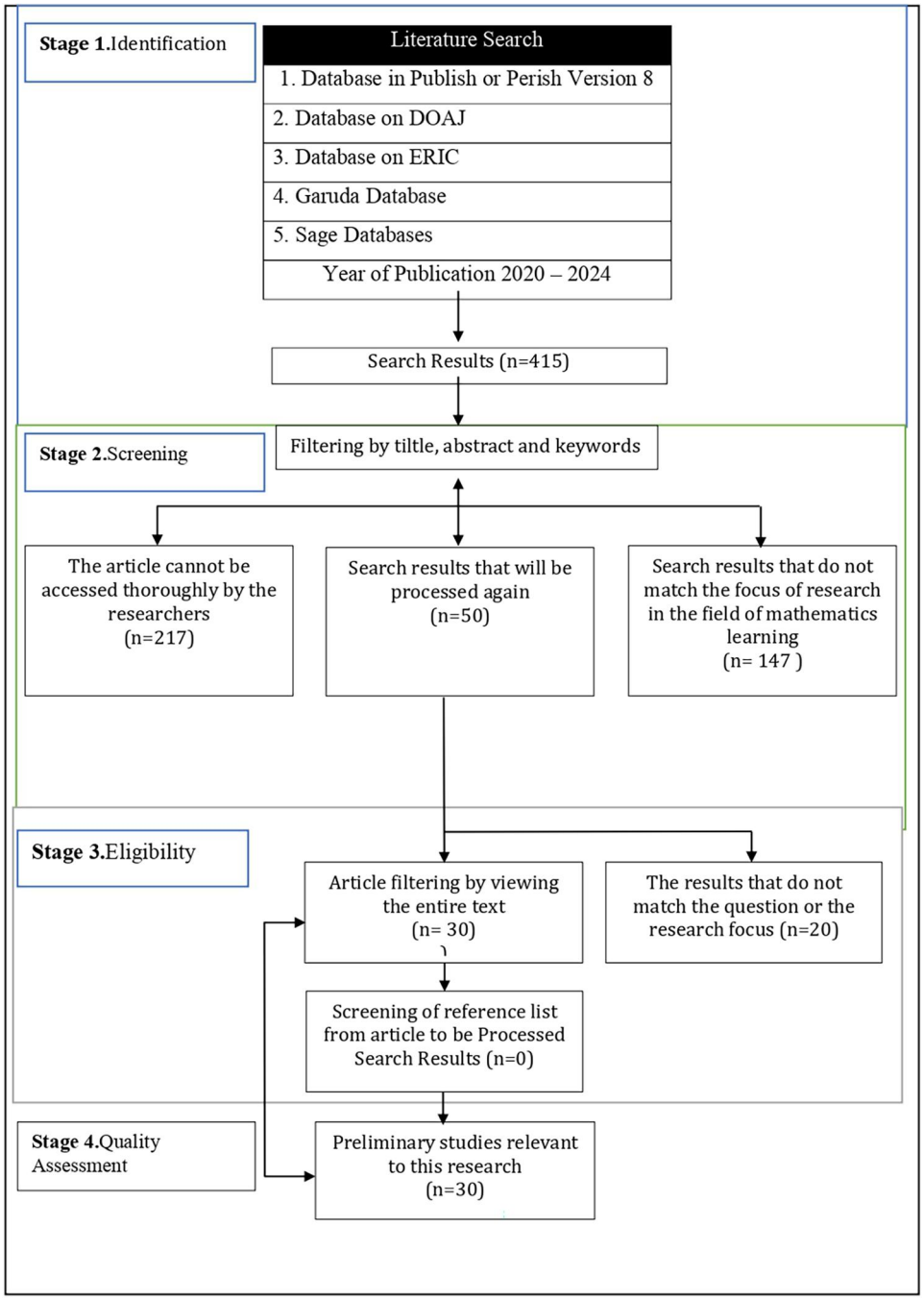


FIGURE 1
Stages of the search and selection of papers (PRISMA flow chart diagram).

217 articles could not be accessed thoroughly due to paywall restrictions, broken links, or lack of full-text availability. Among the 217 articles, 38 were related to mathematics learning but were excluded due to inaccessibility. In addition, the databases used had overlapping entries. The total number of unique articles across all databases, after deduplication, was 332 articles. The reason for focusing on the field of mathematics learning is because it is perceived as a difficult and boring subject (Machaba, 2018). The presence of joyful learning is therefore considered as a good solution that can support students in understanding mathematics in a non-threatening environment (Jeet and Pant, 2023; Shi et al., 2019; Zhang et al., 2021).

3.2.3 Eligibility

At this stage, a usual examination was carried out by evaluating the title and abstract to ensure that all articles met the criteria by looking at the content of the entire text in the

TABLE 1 Technologies integrated into joyful mathematical learning.

Theme	Type of technology	Example tools/platforms	Description of use in studies
Engagement and Assessment	Interactive Quizzes	Wordwall, Quizizz	Used to increase engagement and assess understanding through games
Instructional Support	Video Media	YouTube, Self-recorded Videos	Used for instructional support and self-paced learning
Exploration and Simulation	Game-Based Media	GPS Treasure Hunt, PowerPoint Games	Used for simulation and exploration activities
Student-Centered Learning	Digital Worksheets	Google Forms, Interactive PDFs	Used for student-centered assignments and practice
Modeling and Problem-Solving	Educational Software	GeoGebra, Tarsia Formulator	Used for modeling and problem-solving in geometry and arithmetic
Classroom Interaction	Real-time Feedback Tools	Plickers, Kahoot	Used to provide immediate responses and increase classroom interaction

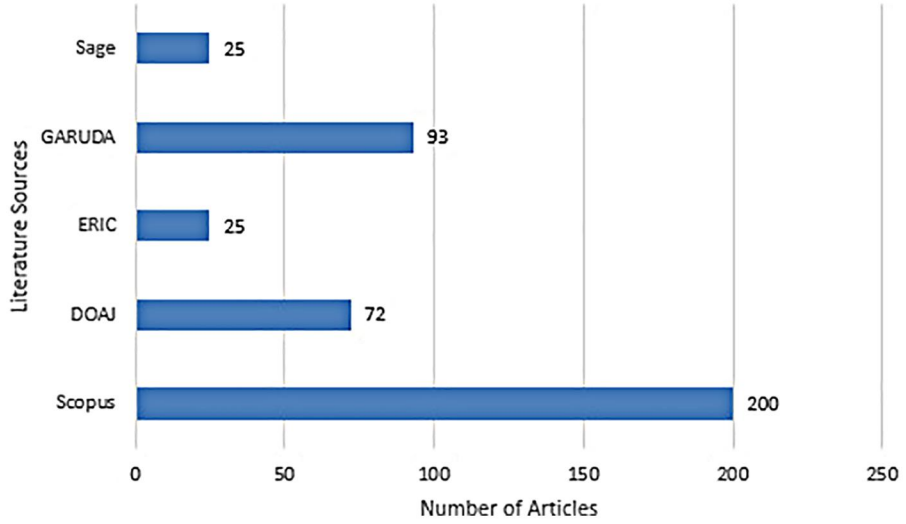


FIGURE 2
Number of articles by database.

article. Some irrelevant articles to the research theme were excluded from the list (20 articles). After examining the eligibility, 30 articles were obtained (Figure 3).

3.2.4 Quality assessment

This stage functioned as a validation process beyond thematic relevance. While the eligibility phase ensured that each article had met the research focus and provided full text access, the quality assessment stage examined the methodological robustness of the studies. This included evaluating the clarity of research design, sample size justification, reliability of instruments, and credibility of findings. Only studies that demonstrated sound methodological quality were retained for the final synthesis. Articles were selected from publications indexed by Scopus, DOAJ, ERIC, Garuda, Sage, or any of the five. We selected articles published during the specified period from 2017 to 2024. Thus, this study reviewed 30 articles, including 11 English and 19 Indonesian-language articles.

3.3 Data presentation and analysis

In presenting the data, the authors conducted a descriptive and thematic analysis by mapping the content of each article to the six research questions. An in-depth reviewing process of each selected article at this stage is crucial to identify which mathematical domains, school levels, and research methods are involved. Pedagogical approaches, technological tools, and findings were discussed. The application of a deductive coding scheme was developed based on the research questions, and categories were refined iteratively during the analysis process. For instance, when categorizing mathematical domains (e.g., geometry, algebra), we coded article content based on explicit mentions of domain-specific instruction or learning objectives. In cases of ambiguity, two researchers independently coded and resolved discrepancies through discussion. Similarly, school-level categories were aligned with the national curriculum stages, while methods were classified using established educational research paradigms.

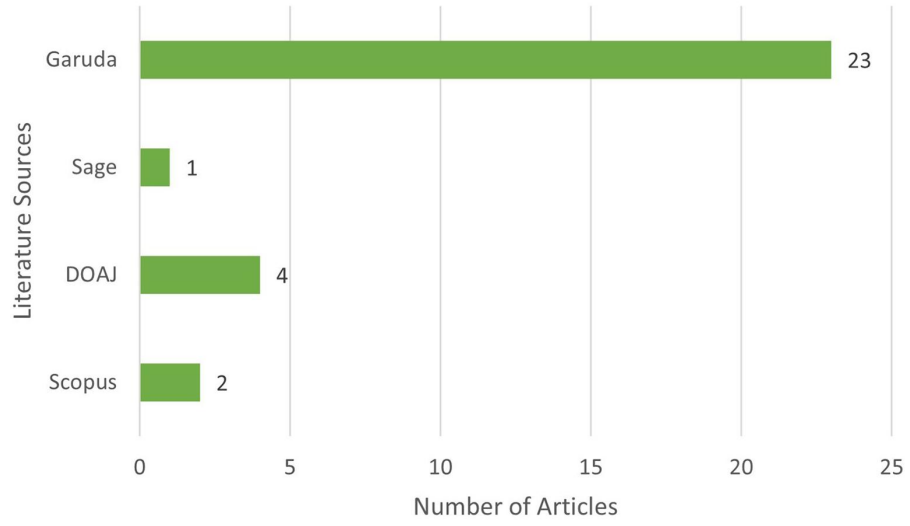


FIGURE 3
Literature frequency.

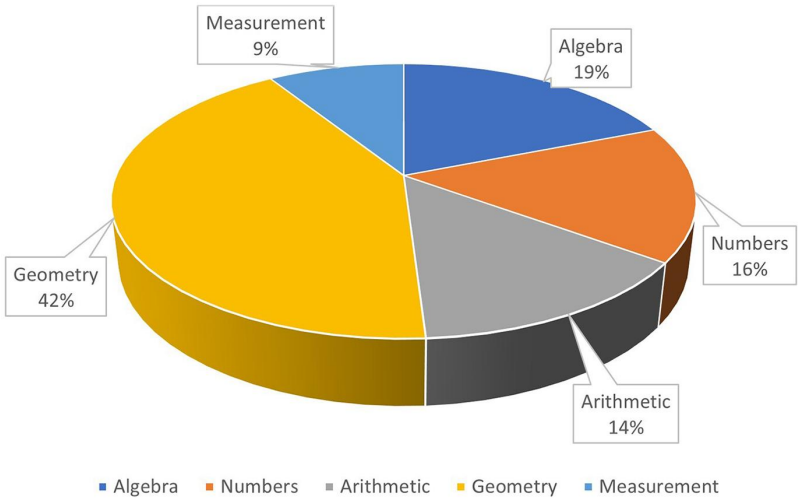


FIGURE 4
Distribution based on mathematics domains taught.

The bibliometric analysis was conducted using Biblioshiny, an R-based interface to the Bibliometrix package (Aria and Cuccurullo, 2017). This tool enabled the authors to visualize keyword co-occurrence, article distribution by journal, and thematic evolution over time. Biblioshiny performs text mining on metadata (titles, abstracts, and keywords) and uses co-word and co-citation analysis to identify prominent research themes. This multi-layered approach ensured that both quantitative trends and qualitative patterns could be captured in alignment with the study's objectives.

4 Results

This research and analysis of joyful learning was conducted by classifying joyful learning based on the domain of mathematics, research method, and school level. Then, the results are presented based on technological involvement, pedagogical approaches, and the study's main findings.

4.1 Description and distribution by domain of mathematics

One of the ways to analyze this study is by examining the domain of mathematics, where joyful learning is applied. Figure 4 presents the distribution of mathematics learning domains that apply joyful mathematical learning. From the 30 articles analyzed, it was found that there were domains where joyful learning was most often applied, namely the geometry domain.

The geometry domain is the highest mathematics domain, with a percentage of 42%. It is one of the topics in mathematics

that is often chosen for the application of a joyful learning approach in primary schools (Kuzle, 2023). It also aligns with what Massarwe (2023) explained: geometry's unique characteristics can facilitate joyful learning through direct interaction and concrete visualization. However, the effectiveness of joyful learning in geometry is not merely due to the tangibility of shapes but also because such representations support playful and meaningful learning experiences aligned with students' developmental levels. In this study, joyful learning is defined as a learning approach that intentionally integrates emotionally positive, cognitively engaging, and socially interactive strategies to promote deeper understanding, such as games, storytelling, music, movement, or culturally contextualized materials. Thus, while hands-on activities like manipulating geometric models may naturally lend themselves to joyful learning, other mathematical domains may require different joyful strategies, such as storytelling in arithmetic or gamified quizzes in algebra, which are tailored to the content and learning objectives. Therefore, joyful learning in the geometry domain does not solely rely on object interaction but also on the pedagogical design that leverages visual, interactive, and emotional engagement to improve conceptual understanding.

Joyful learning in the geometry domain can also be facilitated through the use of instructional media. For example, Qomaria (2021) introduced the concept of geometry using the Tarsia Puzzle, a game-based tool designed to implement fun mathematics learning. The study reported that 83.4% of respondents believed the use of the Tarsia Puzzle could increase student excitement. While these findings reflect positive perceptions of the strategy, they represent expectations about its potential rather than direct evidence of students' learning experiences or outcomes. Nevertheless, such results provide initial support for integrating joyful learning approaches, like the Tarsia Puzzle, into geometry instruction as a way to foster student engagement.

In addition to the geometry domain, mathematics learning in the domains of algebra (Brizuela and Strachota, 2024), numbers (Harefa, 2023), and arithmetic (Hardini et al., 2021) is often applied in joyful learning because of its basic properties that are easy to manipulate and explore creatively. Algebra, numbers, and arithmetic are closely related to patterns, rules, and calculations that can be used as interesting math games.

4.2 Description and distribution by research method

The research method used in the study includes the type of research, specific design, sample, data collection, and data analysis (Lê and Schmid, 2022). Figure 5 presents the distribution of research methods used in the 30 articles.

The classroom action research method (67%) is the most widely used in the analyzed studies, followed by the quantitative method (17%). Classroom action research is the most frequently used research in the study of joyful learning in mathematics. For example, Sutria and Hidayat's study (2019) conveyed that one of the reasons classroom action research is used in the study of joyful learning is to see how the classroom environment is formed by the application of joyful learning and its results on improving student understanding (Ashfaq, 2020; Mooduto and Musa, 2021). This also corresponds to the opinion expressed by Wei et al. (2011), "joyful description as a kind of feeling, expressing and causing great pleasure," and "joyful learning as a kind of learning process or experience which could make learners feel pleasure in a learning scenario or process."

Some of the studies analyzed also used the research and development method, for example, in Utaminingsih's study (2024), which developed interactive digital worksheets, and Sa'adah's study (2023), which developed student worksheets. The research was based on the fact that some students tend to consider mathematics as a complicated subject (Gunawan et al., 2023; Lely et al., 2020), and alternative strategies are needed to create fun learning activities (Conklin, 2014; Utaminingsih, 2024; Waterworth, 2020). On the other hand, qualitative methods were also employed to explain the outcomes of joyful learning. For instance, Thi Thu et al. (2023) described students' experiences and emotions by observing their participation in classroom activities, evaluating the learning products they created, and gathering teachers' feedback on the organization of learning tasks. The results indicated that students initially

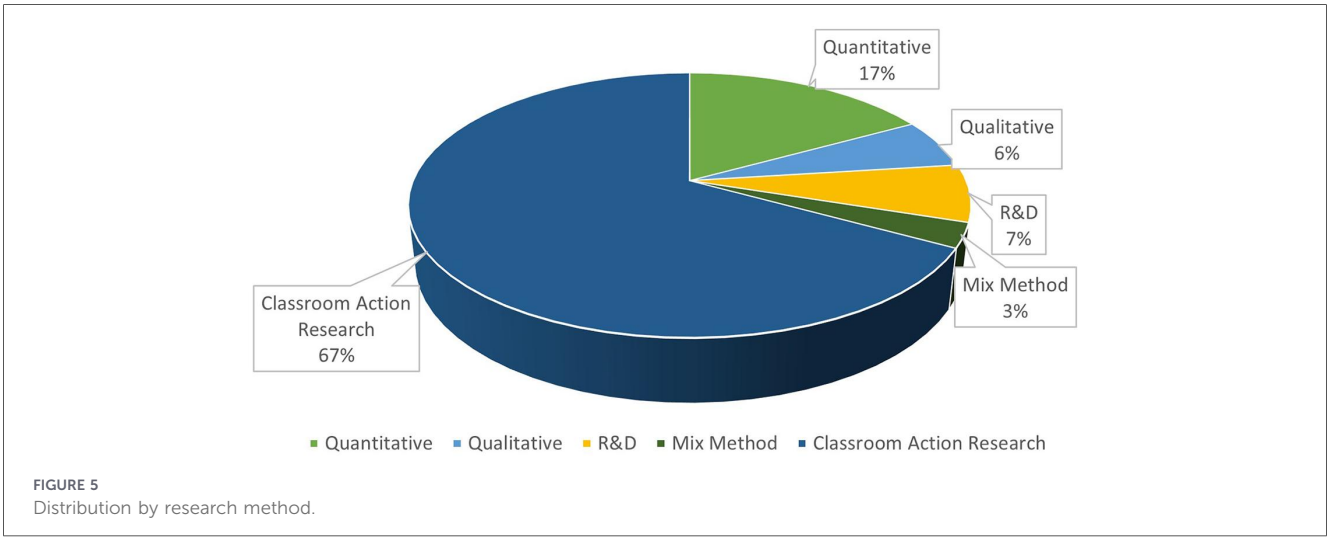


FIGURE 5
Distribution by research method.

participated actively and with enjoyment, collaborated readily to complete assignments, and demonstrated creativity through a variety of unique ideas.

Among the studies that have been conducted, some studies use mixed methods and try to integrate technology in implementing joyful learning in mathematics learning. For example, a study by Chou (2022) tried to combine the use of Plickers through two case scenarios. The first scenario used experimental studies comparing Plickers with traditional tests. At the same time, the second scenario uses a quasi-experimental pretest and posttest design. Then, the study's results were described through qualitative methods by exploring students' perceptions of the two scenarios. From the study results, it was found that fun learning and active engagement during Plickers were indicated to be good and could be adopted as an assessment tool. This corresponds to the opinion of Öner (2023) that the use and integration of technology in the study of joyful learning in mathematics allows teachers to evaluate and optimize their teaching practices in a dynamic and student-centered classroom environment.

4.3 Description and distribution based on the school level of joyful mathematical learning

The application of joyful learning in mathematics learning across the articles analyzed is presented by grade level and school. The integration of joyful learning is predominantly in junior high schools rather than primary schools (Figure 6).

According to the National Ministry of Education Regulation No. 22 of 2006 (Kemdikbud, 2006) on Content Standards, mathematical subjects should be given to all elementary and secondary school students to equip students with logical, analytical, systematic, critical, and creative thinking skills, as well as collaborative skills. The chart describes the distribution of research articles at different school levels. The data showed that "Grade 7" had the largest proportion, 26% of the articles analyzed. This percentage indicates that joyful learning is the most frequently implemented in mathematics education at the seventh-grade level. This trend may reflect the transitional nature of the grade, where students shift from foundational mathematical concepts in primary school to more complex content in secondary school. The use of joyful learning approaches at this level therefore, may be an effective strategy to ease students' cognitive and emotional transition by providing engaging and supportive learning environments, rather than suggesting that this grade yields the highest enjoyment level compared to others. This level is a transition from basic mathematical concepts to more complex ones. One reason to apply joyful mathematical learning was to provide a pleasant learning experience to improve mathematics learning results (Istiqomah and Prihatnani, 2019).

A study by Sidi and Yunianta (2018) showed an improvement in student learning outcomes by applying joyful learning strategies. The results are similar to those obtained by Mujiono (2020), who found that implementing joyful learning strategies improved learning outcomes in grade 7. In addition to grade 7, joyful learning was also observed in studies at the eighth and ninth grade levels, each comprising 6% of the analyzed articles. Although fewer in number, these studies reported similar positive impacts on student learning outcomes, such as the enhancement of engagement and understanding, aligning with the findings from the grade 7 implementations. For example, a study conducted by Khomsah (2022) and Sugyarto (2022) obtained the results that there was an improvement in a large line of the learning outcomes of individuals after performing joyful learning actions. The application of joyful learning in high school is because the material presented is more complicated at the previous level, and its learning is more emphasized in problem-solving efforts (Mujiono, 2020). Meanwhile, some students think mathematics is difficult to learn and understand (Chang, 2013; Minarni and Napitupulu, 2017). So, it can be concluded that joyful learning strategies are necessary in improving student mathematics learning outcomes.

At the primary school level, grade 5 achieved the highest percentage, 23%. Yulidaspa and Mustika's study (2023) found

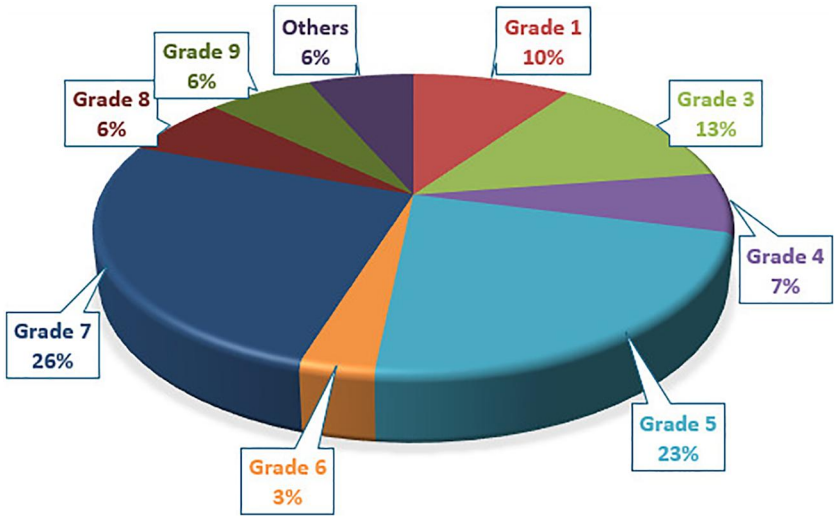


FIGURE 6
Application of joyful learning at the school level.

that learning geometry using a joyful learning model with interactive PowerPoint media has a good influence on improving the learning outcome of the pupils, as the pupils can improve understanding, sharpen memory, and influence the learning results. In the study of Sutria and Hidayat (2019), the learning strategy of joyful learning influenced student learning outcomes in algebraic material and geometry in class 5. According to a study by Sutini (2022), the joyful learning approach can improve students' mathematical learning activity of finding the cube and rectangular prism volume. Interesting conclusions were obtained that joyful mathematical learning implementation in elementary and lower secondary school tends to be done on geometry (Khomsah, 2022; Sutini, 2022; Yulidaspa and Mustika, 2023) and algebra (Haryanti, 2021; Mujiono, 2020; Sugyarto, 2022), although some have applied it to numbers (Sardin et al., 2022) and arithmetic (Merdekawati, 2022) (Figure 5).

The interesting thing about the analysis of this article is that there are some applications of joyful learning done at levels other than elementary and primary secondary schools. For example, a study by Gil-Clemente et al. (2024) investigated the application of joyful learning to mathematical learning for Down syndrome students aged 3–17. On the other hand, the application is not only applied to primary school and secondary school students but can also be applied to teachers. This is in line with the opinion of Nurhajati et al. (2024) that joyful learning applies to students and teachers in teaching-learning with pleasure.

4.4 Description and distribution by integrating technology into joyful mathematical learning

The application of joyful learning in mathematical learning in the article analyzed is presented based on the technology used. More than 35% use some technology that includes video learning and games (Table 1).

The implementation of joyful learning in mathematics involves various technological tools and platforms. The "others" category represents the largest proportion (37%) of technologies used. To provide more clarity, we present a table (Table 1) that categorizes the specific technologies integrated into joyful mathematical learning. Furthermore, this table illustrates how digital tools such as *Wordwall, GeoGebra, GPS-based treasure hunt games, Plickers*, *PowerPoint*-based games, and video recordings are applied to promote engaging and joyful participation in mathematics classrooms. The distribution in Table 1 shows that joyful mathematical learning is supported by various technologies, depending on the teaching and research design. This indicates that joyful mathematical learning is not limited to just one technology. For example, a study by Andriani et al. (2018) that integrated Cooperative Integrated Reading and Composition (CIRC) showed that CIRC influenced the activity, interest, and ability of mathematical reasoning. Merdekawati (2022) conducted a study using social media integration like WhatsApp, YouTube, and online student worksheets. However, some use the latest integrations that are rarely used, such as Plickers (Chou, 2022) and GPS-based Treasure Game technology (Kasmawati, 2020).

Based on the synthesis of the reviewed studies depicted in Figure 7. It shows the variation of technology integration in joyful mathematical learning. Student worksheets occupy the second position most frequently used after the "others" category, with a proportion of 28% of the total articles analysed. These findings indicate that worksheets, especially in digital form, play a role in supporting joyful mathematical learning. For example Dewi et al. (2022) found that students actively engaged in learning tasks even in online learning by integrating worksheets thru Google Forms. Similarly, studies by Utaminingsih (2024) and developed digital worksheets by combining problem-solving tasks and technology, which were able to make students more engaged. It also showed that 87.5% of the activities were effective in mathematics learning. Digital quizzes are the next integration of technology with a proportion of 19%. For example, Sa'adah et al. (2023), uses the *Wordwall* website to create interactive quizzes that are often used to measure student understanding in real-time. Betty's study (2023) uses an interactive quiz that allows students to learn more competently and enjoyably, increasing their motivation to learn mathematics (Chou, 2022). Worksheets allow students to actively engage in the learning process, solve problems, and explore mathematical concepts in an interactive way (Mujiono, 2020).

4.5 Description and distribution by pedagogical approaches of joyful mathematical learning

The application of joyful learning in mathematical learning throughout the analyzed article is presented based on Pedagogical Approaches. Figure 8 shows that the highest of the pedagogical approaches is STEAM (Science, Technology, Engineering, Arts, and Mathematics) with 28%. This indicates that STEAM is the most frequently found pedagogical approach in the articles analyzed. STEAM is an interdisciplinary approach integrating different fields of study to provide a more holistic and contextual learning experience (Widyawulandari et al., 2019). In the context of mathematics-focused research, STEAM was typically employed to situate mathematical concepts within broader, real-world problems that also involved elements of science, technology, engineering, or the arts. Thus, while mathematics remained the primary subject of instruction, it was taught in connection with other disciplines to enhance relevance, creativity, and student interest (Kruk et al., 2022).

Several studies further illustrate how STEAM has been combined with additional theoretical foundations to support joyful learning. For instance, Thu et al. (2023) employed the 5E instructional model (Engage, Explore, Explain, Elaborate, Evaluate) within a STEAM framework and reported that students participated actively and with enjoyment, collaborated effectively in group tasks, and expressed creativity through diverse and original ideas. Similarly, Lavy's study (2023) integrated Cognitive Load Theory with STEAM to ensure that interdisciplinary problem-solving tasks remained cognitively manageable, thereby sustaining student engagement and reducing frustration. Aosi (2019) applied STEM which can be

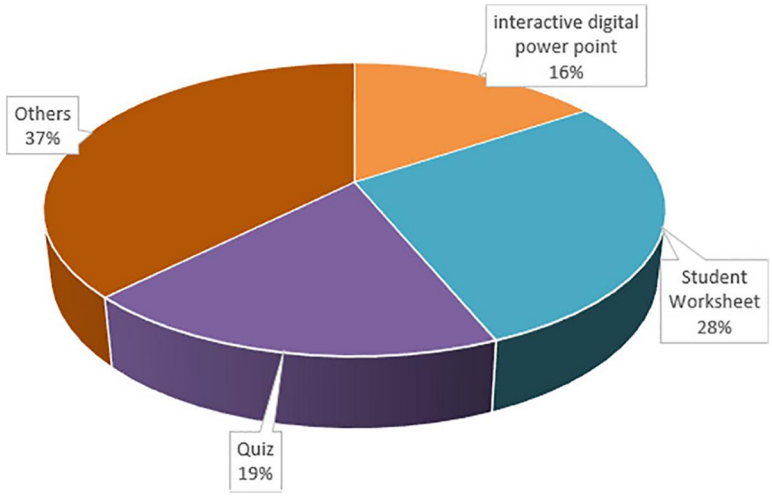


FIGURE 7
Integration of technologies used in joyful mathematical learning.

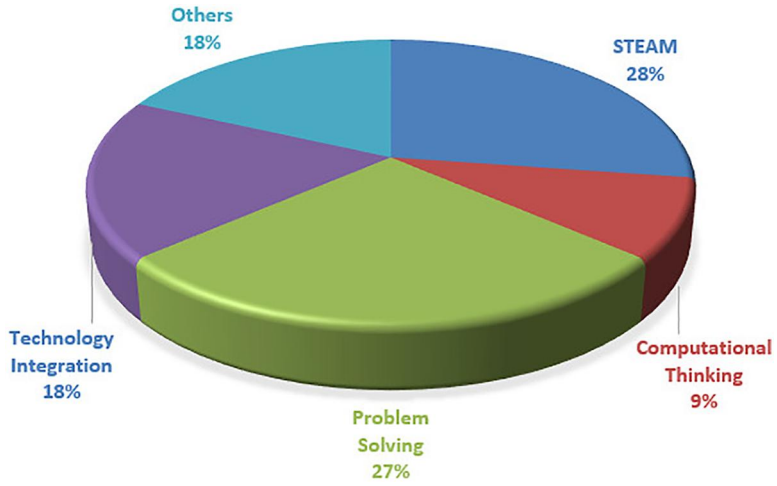


FIGURE 8
Pedagogical approaches used in joyful mathematical learning.

seen as a subset of STEAM (Kang, 2019), to focus more narrowly on mathematical and scientific reasoning. These examples demonstrate that STEAM, when combined with other theoretical perspectives, can create joyful and meaningful learning experiences by balancing interdisciplinary exploration with cognitive support for learners. Problem-solving is the second most widely used pedagogical approaches in the article analyzed with 27%. This approach emphasizes the ability of students to solve mathematical problems independently, using analytical and logical skills (Mishra, 2017). The integration of joyful learning with a problem-solving approach impacts students by encouraging active engagement in learning, exploratory thinking, and consistency during task completion (Istiqomah and Prihatnani, 2019; Mishra, 2017). Chen et al. (2016) stated that in joyful learning, problem-solving helps students engage actively in the learning process, where they are encouraged to find solutions through exploration and experimentation. For example, a study conducted by Hidayati et al. (2022) showed that joyful learning in mathematical learning helps students with problem-solving skills.

Technology integration gains 18% of the pedagogical approaches used. According to Shen and Ho (2020), it is a teaching method that uses digital technology to improve the learning-teaching process. This approach includes using various digital devices and applications required in mathematics learning (Shatri, 2020). In the article that has been analyzed, it has been found that there is an approach to learning through technology, for example, in the study of Merdekawati (2022), which uses the Geogebra Applet as the use of technology in learning mathematics. The research concluded that activities in the GeoGebra applet empowered students to develop their understanding of integer operations. Data shows that students can start exercises and continue to try activities when they encounter difficulties or get the latest ideas. GeoGebra activities

TABLE 2 Important findings from joyful mathematical learning.

No	Name	Title	Results
1	Thu et al. (2023)	Applying the Engage, Explore, Explain, Elaborate, and Evaluate Procedure in STEAM Education for Primary Students: A Sample with the Topic "My Green Garden"	Students actively participate in learning activities with a sense of joy, ready to work together to complete assigned tasks. The students' creativity is evident through the variety of unique ideas they present.
2	Gil-Clemente et al. (2024)	Involving Parents and Family Environment for Enhancing the Mathematics Education of Children with Down Syndrome	Videos open the way to online learning for children with Down syndrome.
3	Lavy (2023)	Leveraging the Pied Piper Effect: The Case of Teaching Programming to Sixth-Grade Students Via Music	Melodycode changes students' stereotypes about programming. Stereotypes such as programming being a male profession that requires good math and other skills.
4	Utaminingsih (2024)	Management of Mathematics Learning Based on Interactive Digital Worksheets to Improve Students' Critical Thinking Ability	The questionnaire results showed students' need for digital-based interactive learning media on fractional number material. 87.5% of students agree with the teacher's assumption of implementing digital-based interactive learning media.
5	Sa'adah et al. (2023)	Developing Student Worksheets Based on Joyful Learning on Numbers and Arithmetic Operations for First Grade Elementary School	Produced teaching materials, namely mathematics student worksheets for first-grade students on counting and arithmetic operations up to 10
6	Habeahan et al. (2023)	The Effect of the Joyful Learning Method on the Third-Grade Students' Learning Outcomes in Mathematics	This research found that the use of joyful learning methods has a significant positive influence on student learning outcomes.
7	Kasmawati (2020)	Implementation of the Joyful Learning Model Assisted by GPS *Harta Karun* to Increase Student Learning Motivation in Thematic Learning at SD Negeri 2 Lamokato	The joyful learning model assisted by GPS treasure can increase students' learning motivation in the fifth grade's thematic subjects. The joyful learning model that can increase students' learning motivation in this research is joyful learning assisted by "GPS Harta Karun" (PowerPoint spin game in cycle and treasure hunt game).
8	Qomaria (2021)	Teachers' Perception Towards the Use of Tarsia Puzzle to Create Joyful Learning of Mathematics	This research shows that elementary school teachers perceive tarsia puzzles well in mathematics learning.
9	Sardin et al. (2022)	The Effect of Card Media in the Joyful Learning Model to Develop Fifth Grade Students' Conceptual Understanding of Fraction at SDN 1 Masiri	The joyful learning model significantly influences students' understanding of mathematical concepts.
10	Sutini (2022)	Joyful Learning Approach to Increase 3D Shape Learning Activities for Fifth-Grade Students of SD Negeri 6 Gedong Tataan Pesawarant Regency 2019/2020	A joyful learning approach can increase students' mathematics learning activities by determining the volume of cubes and rectangular prisms.
11	Hatmawati (2021)	The Effect of Implementing the Joyful Learning Strategy on Mathematics Learning Outcomes in 2D Shapes Material for Third Grade Elementary School Inpres 130 Tarowang, Jeneponto Regency	After implementing the joyful learning strategy, there was a significant increase in students' mathematics learning outcomes. The average score increased from 53.33 on the pretest to 85.00 on the posttest. Apart from that, there was an increase in the learning outcomes category from medium to high.
12	Hidayati et al. (2022)	The Implementation of Joyful Learning Strategies in Fifth Grade Students' Mathematics Learning Outcomes at SDN 5 Karang Anyar	A good joyful learning strategy in learning to improve fifth-grade mathematical problem-solving is included in activities that are right on target
13	Chou (2022)	Using Plickers to Support Student Learning in Rural Schools: A Comprehensive Analysis	This study investigated the instructional effectiveness of Plickers in two different disciplines. The research results in the English classroom scenario confirmed that students' overall attitudes toward Plickers remained positive. In addition, students' perceptions of the integration of clickers emphasize fun learning, active involvement, and motivated learning. Research results in the mathematics classroom scenario show that Plickers can enable students to obtain better performance and learning improvements. Students' perceptions of *Plickers* are similar to those of English class scenarios.

(*Continued*)

TABLE 2 Continued

No	Name	Title	Results
14	Mujiono (2020)	Joyful Learning to Improve Junior High School Students' Algebra Learning Achievement in SMP Negeri 1 Wonomerto Class	It can be seen that student learning outcomes in mathematics learning algebra material using the joyful learning strategy from pre-cycle to cycle I have increased. Data obtained from test results show that student learning outcomes have increased gradually and are quite good compared to before the joyful learning strategy was implemented.
15	Sugyarto (2022)	The Effect of Joyful Learning Strategies on the Results of Learning to Understand Quadratic Functions	Student learning outcomes and engagement increased with the application of joyful learning in quadratic functions learning.
16	Haryanti (2021)	Joyful Learning Strategy in Efforts to Improve Student Learning Outcomes in Mathematics Subjects in Class VII Middle Schools	Student learning outcomes increased from pre-cycle (26.41) to cycle 1 (77.41).
17	Merdekawati (2022)	Enhancing Students' Understanding of Operation in Integers Through Joyful Learning with Geogebra^{TM} Apple	Students can communicate and interact with each other and with teachers through the application. Students work on online activities as occurred during the Covid-19 pandemic period
18	Khomsah (2022)	Think Square Share (TSS) Structural Approach to Improve Active, Joyful, Effective Learning (AJEL) on Curved Side 3D Shapes	This research shows that based on the results of student evaluation tests in cycle I, the class average score was 73.39 with a completion percentage of 64.51% in the medium category, in cycle II the class average score was 81.13 with a completion percentage of 70.97%. high category, and cycle III obtained an average class score of 62.58 with a completion percentage of 41.97% in the low category.
19	Hatmawati (2021)	The Effect of Implementing Joyful Learning Learning Strategies on Mathematics Learning Outcomes in Flat Building Material for Third-Grade Elementary School Inpres 130 Tarowang, Jeneponto Regency	There is an influence of the application of joyful learning strategies on mathematics learning outcomes of third grade students.
20	Dewi et al. (2022)	The Implementation of Joyful Learning to Improve Learning Outcomes of Integers for Seventh Grades of VII SMP Negeri 3 Pati	The application of the joyful learning method can improve the learning outcomes of seventh grade students on learning integer material. This is proven by an increase in the average and percentage of students who have completed the minimum completed standard.
21	Betty (2023)	Efforts to Increase Learning Motivation of Mathematics through Joyful Learning Methods in the Seventh Grades at MTSN I Palembang Students	Applying the joyful learning strategy to several materials can make students active in learning and make it easier for educators to deliver material and teaching materials.
22	Yulidaspa and Mustika (2023)	The Effect of the Joyful Learning Model Using Interactive Power Point Media on the Learning Outcomes of Fifth Grade Students at SD Negeri 193 Pekanbaru	There is an influence on student learning outcomes by using the joyful learning model with interactive Power Point media on the learning outcomes of fifth grade students
23	Wahyudin et al. (2023)	Efforts to Increase Learning Concentration through Joyful Learning Methods Based on Ice Breaking at Sdn 4 Dongos	The application of the ice breaking based joyful learning method which focuses on learning that feels fun, light and does not make students burdened
24	Sari (2023)	Improving Mathematics Learning Outcomes in Algebra Material Using Joyful Learning Learning Strategies at SMP Negeri 4 Kotanopan	The implementation of the joyful learning strategy shows an increase in algebra learning outcomes of seventh grade students
25	Aosi (2019)	STEM Based Learning to Create Joyful Investigating Pumpkins for Mathematical Learning in Primary School	It can be concluded from this activity that the STEM approach has been proven to improve learning outcomes and critical thinking of first grade students
26	Istiqomah and Prihatnani (2019)	Improving Student Learning Outcomes and Attitudes towards Joyful Mathematics Learning	The application of joyful learning strategies will create an environment that is relaxed, interesting, and does not make students hesitate to try, as well as challenging learning situations in the learning process.
27	Sutria and Hidayat (2019)	The Effect of ICT-Assisted Joyful Learning Strategies on Student Learning Outcomes	The learning outcomes of students who use the joyful learning strategy on fifth grade student learning outcomes on the main material of numbers and flat shapes with an average score of 73.29.

(*Continued*)

TABLE 2 Continued

No	Name	Title	Results
28	Yusra (2018)	Profile of Mathematical Communication Skills with Joyful Learning Based on Malay Cultural Context	Learning using joyful learning based on Malay cultural contexts can help students discover mathematical concepts independently in an atmosphere that they enjoy, can make them more active in asking questions and have the ability to think more imaginatively and variedly in solving mathematical problems which makes them better able to communicate mathematics better.
29	Sidi and Yunianta (2018)	Improving Seventh Grade Students' Mathematics Learning Outcomes of Algebra Using Joyful Learning Strategies	The implementation of the joyful learning strategy shows an increase in mathematics learning outcomes of seventh grade students
30	Andriani et al. (2018)	The Effectiveness of the Cooperative Integrated Reading and Composition Learning Model with Joyful Learning Approach to Seventh Students' Mathematical Reasoning Ability on Pythagorean Theorem	It was concluded that mathematics learning using the cooperative integrated reading and composition learning model, a joyful learning approach to mathematical reasoning abilities, was effective.

encourage students to experiment with mathematics concepts in an interactive and communicative environment with other students and teachers and enable them to reflect and draw conclusions at the end of the activity. Tarsia Formulator software (Qomaria, 2021) and Plickers software (Chou, 2022) are also commonly used. All these studies show that learning through technology helps students improve their mathematical abilities and visualize them (Attard and Holmes, 2022).

In general, research on joyful learning provides information that significant changes have taken place and significantly impacted mathematical learning. Based on the explanation of results and data analysis, it is seen that applications of joyful mathematical learning vary in methods, approaches, and media used at different levels of education. Most studies indicate that the use of a variety of ways, such as security tools and educational games, for example, the use of video recording (Hardini et al., 2021; Khotimah et al., 2023) and social media (Andriani et al., 2018; Hardini et al., 2021) are the most frequently applied strategies in joyful learning. This shows that interactive learning involving students is preferred and effective in improving mathematical understanding.

In addition, it is important to note that variations in this approach are based not only on curriculum needs but also on individual differences among students. Some studies also emphasize the importance of the pedagogical approach adjustment (Figure 8) used in the study based on the level of material difficulty and student characteristics, which suggests a realization that there is no single approach that fits all situations. This synthesis underlines the importance of flexibility in applying joyful learning to suit each student group's context and specific needs. The findings of joyful learning studies from 30 articles are presented in Table 2.

In general, research on joyful learning provides information that there is an increase in learning outcomes, understanding, and motivation in implementing mathematics learning (Conklin, 2014; Cronqvist, 2021). Increasing understanding, knowledge, motivation, and learning outcomes are carried out by applying learning methods and approaches, connecting them with games, media, tools, and technology to the mathematics domain that will be taught to students.

6 Discussion and concluding remarks

This study synthesizes recent literature on joyful learning in mathematics, highlighting how it has been conceptualized, implemented, and examined across different school levels and research designs. The findings reveal that joyful learning is frequently integrated into classroom action research, supported by technology, enriched through cultural relevance, and strengthened by interdisciplinary approaches such as STEAM. Across these applications, studies consistently indicate that joyful learning can foster mathematical understanding, enhance student motivation, and increase engagement when adapted to curricular needs and learner characteristics. Empirical evidence further shows its potential to improve learning outcomes, particularly at the elementary and middle school levels, where it creates a more enjoyable classroom atmosphere and reduces student anxiety toward mathematics (Ermawati and Amalia, 2023; Marchy et al., 2022; Turgut and Turgut, 2020). Various strategies such as interactive worksheets (Ramury et al., 2020; Sa'adah et al., 2023), STEAM projects (Aosi, 2019; Kang, 2019; Diego-Mantecon et al., 2021), ice-breaking activities (Wahyudin et al., 2023), and culture-based approaches (Hidayati et al., 2022; Khomsah, 2022) illustrate the flexibility of joyful learning in promoting creativity, collaboration, and critical reasoning in mathematics classrooms.

These results underscore the significance of joyful learning not only as a tool to improve academic outcomes but also as a way to cultivate communication skills, confidence, and positive attitudes toward mathematics. Nevertheless, despite its demonstrated benefits, joyful learning has not yet been widely or systematically implemented in mathematics, particularly at the elementary level. Further research is therefore needed to explore its long-term effects, scalability, and adaptability across diverse mathematical domains and student populations, ensuring that students at an early stage develop strong mathematical foundations through enjoyable and meaningful learning experiences.

Author contributions

ZP: Conceptualization, Funding acquisition, Methodology, Resources, Supervision, Writing – original draft, Writing –

review & editing. JA: Supervision, Validation, Writing – review & editing. RK: Supervision, Validation, Writing – review & editing. YG: Formal analysis, Investigation, Methodology, Visualization, Writing – original draft. DD: Formal analysis, Supervision, Validation, Writing – review & editing. AA: Formal analysis, Methodology, Supervision, Validation, Visualization, Writing – review & editing.

Funding

The author(s) declared that financial support was received for this work and/or its publication. The authors would like to thank KEMENDIKBUDRISTEK and LPPM Universitas Riau for funding this research under the DRTPM research grant number 20685/UN.19.5.1.3/AL.04/2024.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

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